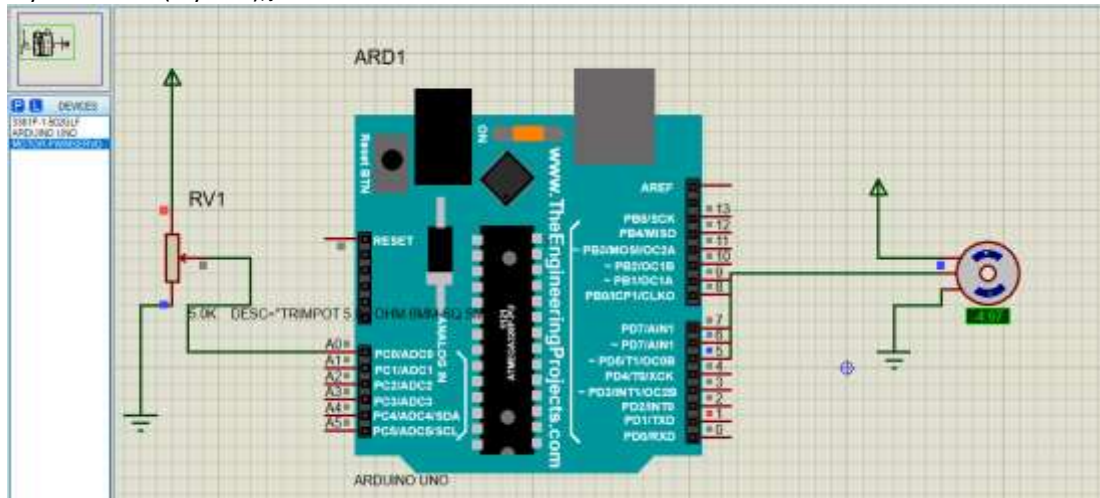


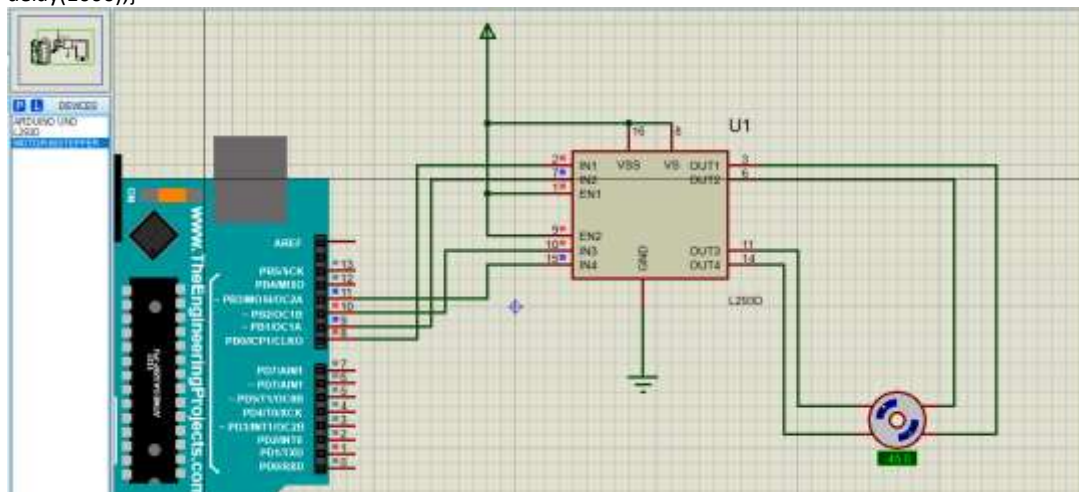
1. POT AND SERVO

```
#include <Servo.h>
Servo myservo;
void setup(){
  myservo.attach(5);
  Serial.begin(9600);
  pinMode(A0,INPUT);}
void loop(){
  int data=analogRead(A0);
  int mydata=map(data,0,1023,0,180);
  myservo.write(mydata);}
```



2. STEPPER

```
#include <Stepper.h>
const int apr =40;
Stepper stepper(apr, 8, 9, 10, 11);
void setup(){
  stepper.setSpeed(10);
  Serial.begin(9600);}
void loop(){
  stepper.step(apr);
  delay(1000);
  stepper.step(-apr);
  delay(1000);}
```



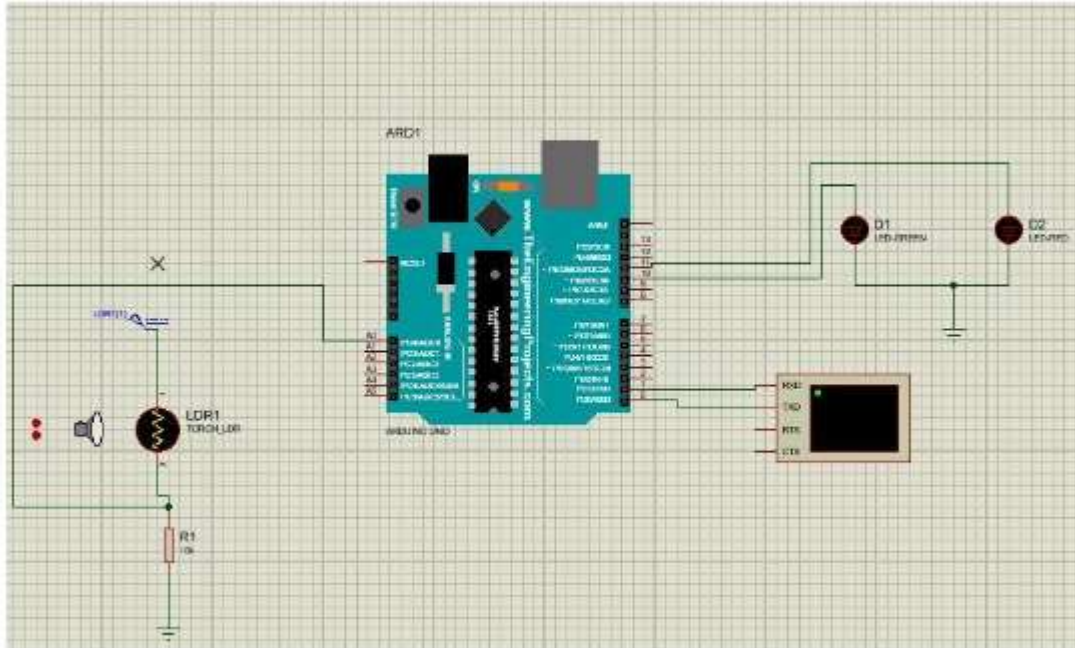
3A ANALOG (LED based on the LDR (Light Dependent Resistor -Analog sensor) sensor)

```
int x;
void setup(){
  pinMode(10,OUTPUT);
  pinMode(11, OUTPUT);}
```

```

Serial.begin(9600);}
void loop(){
  x= analogRead(A0);
  Serial.println(x);
  if (x<500)
  {
    digitalWrite(10,HIGH);
    digitalWrite(11,LOW);
  }
  if (x>500)
  {
    digitalWrite(10,LOW);
    digitalWrite(11,HIGH);
  }
  delay(200);}

```

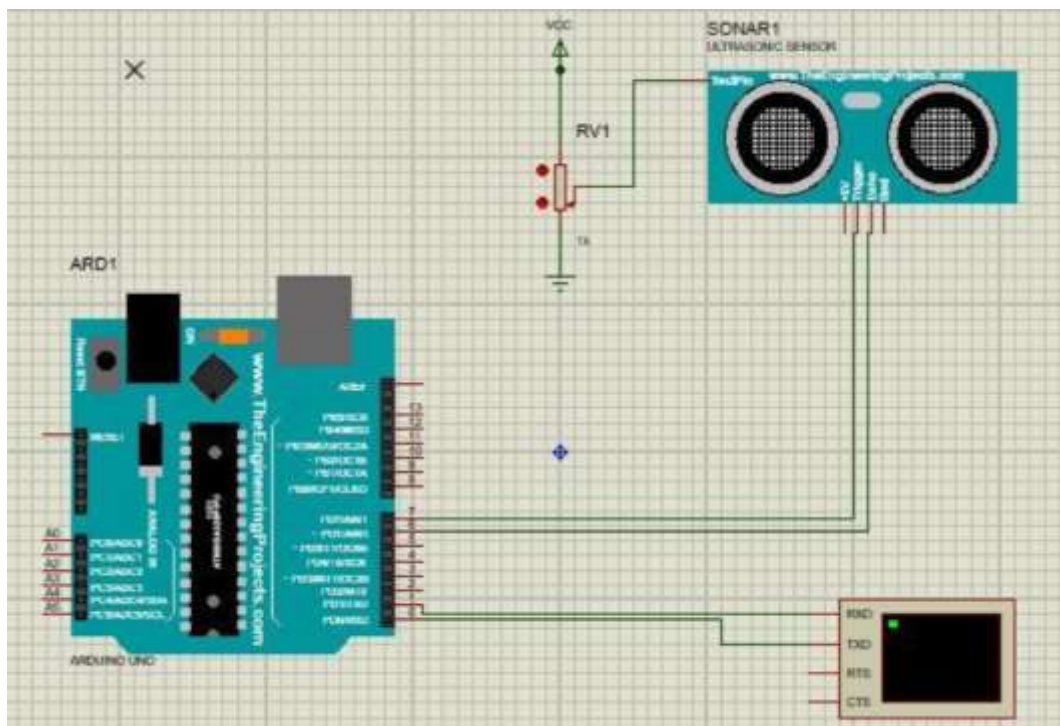


3B DIGITAL (To measure the distance from the target based on the Ultrasonic waves)

```

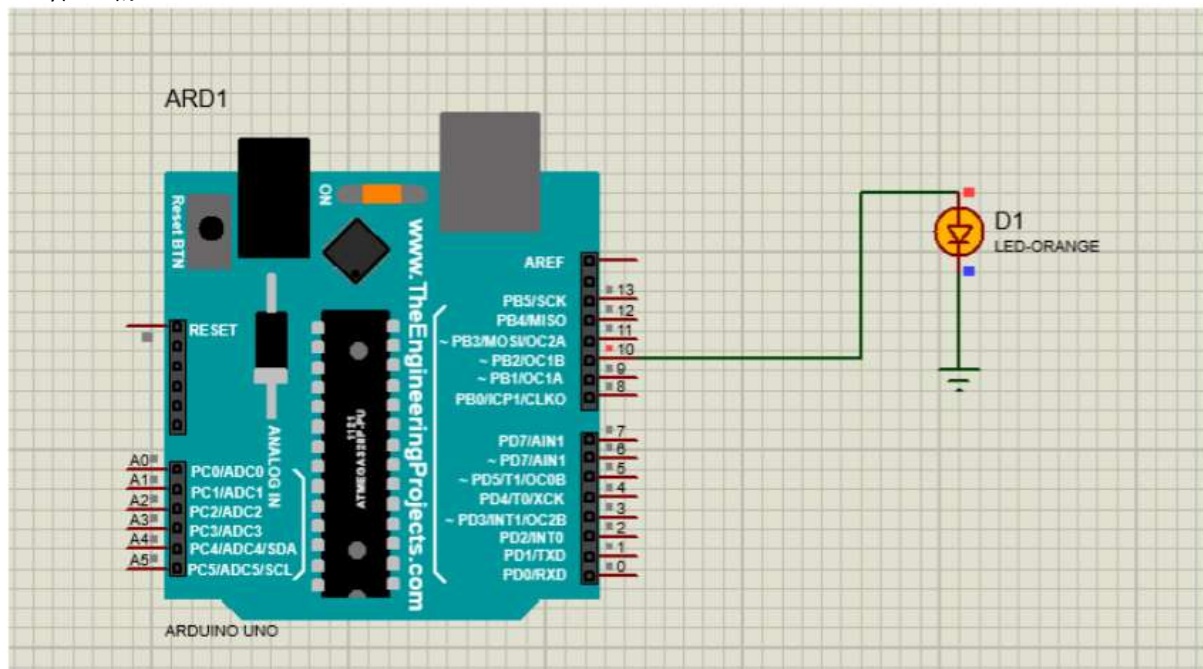
const int pingPin=7;
const int echoPin=6;
void setup(){
  Serial.begin(9600);}
void loop(){
  long duration, inches, cm;
  pinMode(pingPin,OUTPUT);
  digitalWrite(pingPin, LOW);
  delayMicroseconds(2);
  digitalWrite(pingPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(pingPin, LOW);
  pinMode(echoPin, INPUT);
  duration =pulseIn(echoPin,HIGH);
  inches=microsecondsToInches(duration);
  cm=microsecondsToCentimeters(duration);
  Serial.print(inches);
  Serial.print("in, ");
  Serial.print(cm);
  Serial.print("cm");
  Serial.println();
  delay(100);}
long microsecondsToInches(long microseconds){
  return microseconds / 74 / 2;}
long microsecondsToCentimeters(long microseconds){
  return microseconds / 29 / 2;}

```



4. LED / BUZZER

```
const int LED=10;
void setup(){
  pinMode(LED,OUTPUT);}
void loop(){
  digitalWrite(LED,HIGH);
  delay(1000);
  digitalWrite(LED,LOW);
  delay(2000);}
```



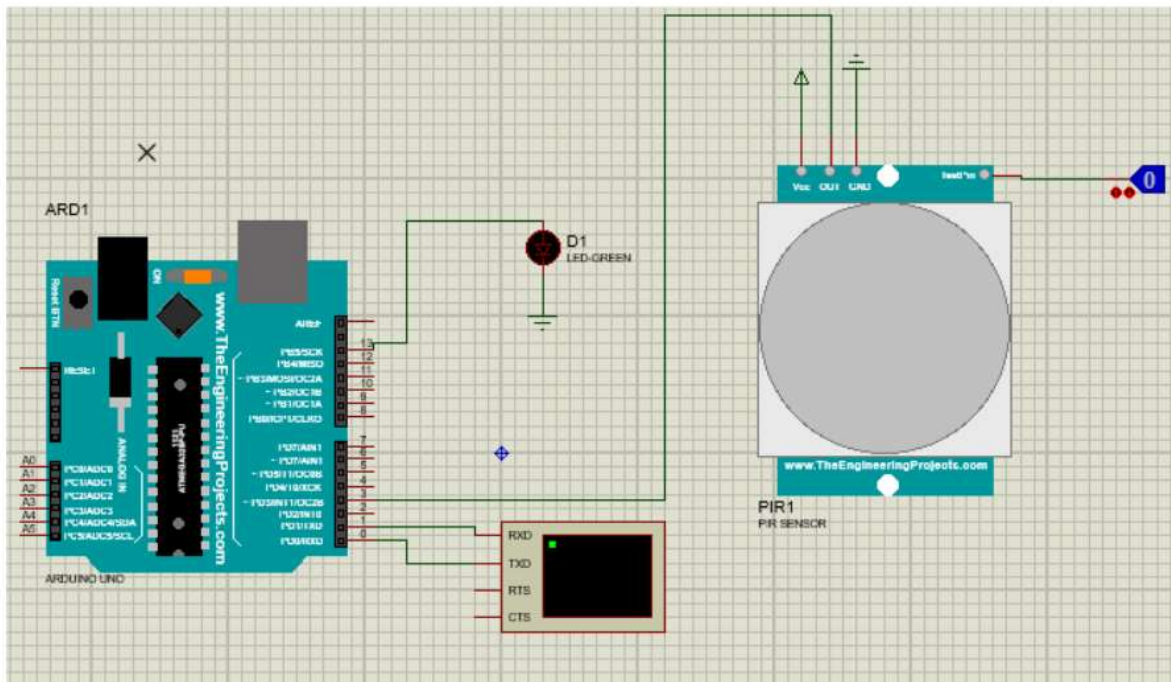
5. PIR

```
const int pirpin=3;
const int ledpin=13;
void setup(){
  pinMode(pirpin,INPUT);
  pinMode(ledpin,OUTPUT);
  Serial.begin(9600);}
void loop(){
  int motionState=digitalRead(pirpin);
```

```

if(motionState==HIGH){
digitalWrite(ledpin, HIGH);
Serial.println(" Motion Detected");}
else{
digitalWrite(ledpin,LOW);
Serial.println("Motion Stopped");}
delay(1000); }

```



6. DHT22

```

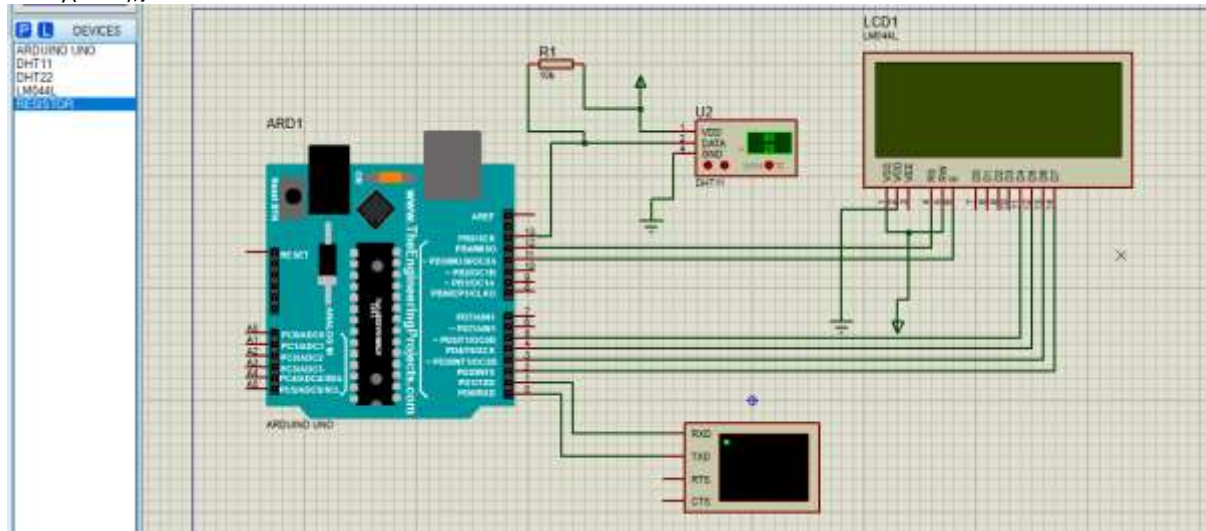
#include <DHT.h>
#include <LiquidCrystal.h>
LiquidCrystal lcd(12, 11, 5, 4, 3, 2); // LCD pin connection (RS, E, D4, D5, D6, D7)
#define DHTPIN 13
#define DHTTYPE DHT22
DHT dht(DHTPIN, DHTTYPE);
void setup() {
  lcd.begin(16, 2);
  Serial.begin(9600);
  dht.begin();
  lcd.print("System Starting");
  delay(2000);
  lcd.clear();}
void loop() {
  float temp = dht.readTemperature();
  float hum = dht.readHumidity();
  if (isnan(temp) || isnan(hum)) {
    lcd.clear();
    lcd.print("Sensor Error");
    Serial.println("Failed to read from DHT22");
    delay(2000);
    return;
  }
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Temp: ");
  lcd.print(temp);
  lcd.print(" C");
  lcd.setCursor(0, 1);
  lcd.print("Hum: ");
  lcd.print(hum);
  lcd.print(" %");
  Serial.print("Temperature: ");
  Serial.print(temp);

```

```

Serial.println(" C");
Serial.print("Humidity: ");
Serial.print(hum);
Serial.println(" %");
delay(2000);}

```



7. FLAME SENSOR

```

#include <LiquidCrystal.h>
LiquidCrystal lcd(13, 12, 11, 10, 9, 8);
const int SensorPin = 2;
const int LEDPin = 7;
void setup()
{
  pinMode(SensorPin, INPUT);
  pinMode(LEDPin, OUTPUT);
  lcd.begin(20, 4);
  lcd.setCursor(0,0);
  lcd.print(" THE BRIGHT LIGHT ");
  lcd.setCursor(0,1);
  lcd.print(" FIRE DETECTION SYS");}
void loop(){
  if(digitalRead(SensorPin) == HIGH){
    digitalWrite(LEDPin,HIGH);
    lcd.setCursor(0,3);
    lcd.print(" Fire Detected ");
    delay(70);
    lcd.setCursor(0,3);
    lcd.print(" Fire Detected. ");
    delay(70);
    lcd.setCursor(0,3);
    lcd.print(" Fire Detected.. ");
    delay(70);
    lcd.setCursor(0,3);
    lcd.print(" Fire Detected... ");
    delay(70);}
  else{
    digitalWrite(LEDPin,LOW);
    lcd.setCursor(0,3);
    lcd.print(" No Fire ");}}

```

